```
import pandas as pd
import matplotlib.pyplot as plt

# Create a sample dataset
data = {
    "Student": ["A", "B", "C", "D", "E", "F", "G",
    "Study_Hours": [1, 2, 3, 4, 5, 6, 7, 8],
    "Scores": [45, 50, 60, 65, 72, 80, 85, 95]
}

df = pd.DataFrame(data)

# 1. Inspect the dataset
print("FIRST FIVE ROWS")
print(df.head())

print("\nDATASET INFORMATION")
print(df.info())

# 2. Check for missing values
print("\nMISSING VALUES")
print(df.isnull().sum())

# 3. Calculate descriptive statistics
print("\nSUMMARY STATISTICS")
print(df[["Study_Hours", "Scores"]].describe())

# 4. Calculate correlation
correlation = df["Study_Hours"].corr(df["Scores"])
print("\nCORRELATION:", round(correlation, 3))

# 5. Visualize the relationship
plt.figure(figsize=(8, 5))
plt.scatter(df["Study_Hours"], df["Scores"])
plt.xlabel("Study Hours")
plt.ylabel("Scores")
plt.title("Study Hours vs Student Scores")
```

```
plt.grid(True)
plt.show()
```

```
FIRST FIVE ROWS
  Student  Study_Hours  Scores
0       A            1      45
1       B            2      50
2       C            3      60
3       D            4      65
4       E            5      72

DATASET INFORMATION
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 3 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Student      8 non-null      object
 1   Study_Hours  8 non-null      int64
```

Gemini

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Student": ["A", "B", "C", "D", "E"],
    "Study_Hours": [2, 4, 3, 5, 6],
    "Scores": [55, 70, 65, 80, 90]
}

df = pd.DataFrame(data)

print("DATASET")
print(df)

print("\nSUMMARY STATISTICS")
print(df.describe())

df.plot(
    x="Study_Hours",
    y="Scores",
    kind="scatter",
    title="Study Hours vs Scores"
)
```

```
plt.show()
```

```
DATASET
  Student  Study_Hours  Scores
0       A            2      55
1       B            4      70
2       C            3      65
3       D            5      80
4       E            6      90

SUMMARY STATISTICS
       Study_Hours     Scores
count     5.000000   5.000000
mean      4.000000  72.000000
std       1.581139  13.509256
min       2.000000  55.000000
25%       3.000000  65.000000
50%       4.000000  70.000000
75%       5.000000  80.000000
max       6.000000  90.000000
```



